

Summary for the Porject

Imbalance-handling: We compared six strategies - class-weighting, random undersampling, SMOTE, ADASYN, SMOTETomek, and SMOTEENN - on the same held-out test set, ranked by PR-AUC. ADASYN scored the highest PR-AUC (0.708) but its precision of only 0.017 meant roughly 98% of its flags would be false alarms, making it impractical. Among the rest, SMOTEENN was strongest (PR-AUC 0.689), narrowly ahead of SMOTETomek and SMOTE (both 0.671), with class-weighting close behind (0.672) and random undersampling weakest (0.600). We selected **SMOTETomek** as our primary balancing strategy for final model training, since as a hybrid method it cleans up noisy overlap at the class boundary while matching SMOTE's precision/recall balance.

Modeling & evaluation: Of the supervised models, **XGBoost** scored the highest PR-AUC (0.818), narrowly ahead of Random Forest (0.795) and well ahead of Logistic Regression (0.671). Both supervised models clearly outperformed the unsupervised detectors: Isolation Forest reached a modest PR-AUC of 0.104, while Local Outlier Factor scored a PR-AUC of just 0.002 and a ROC-AUC of 0.506 - statistically indistinguishable from random guessing on this dataset, and a genuine limitation of that specific configuration rather than a general "unsupervised models are weaker" story. The unsupervised models remain conceptually valuable as a complementary safety net for catching fraud patterns not yet present in historical labels, even though Isolation Forest was the only one to add real signal here.

Threshold choice: Rather than the default 0.5 cutoff, we selected a threshold using a cost sensitive analysis: assuming a missed fraud costs roughly \$500 and a false alarm costs roughly \$10, the cost-minimizing threshold came out to **0.12**, producing 129 false alarms and only 15 missed frauds for a total expected cost of \$8,790. This is well below both the default 0.5 cutoff and the F1-optimal threshold of 0.979, reflecting that catching fraud is worth far more than avoiding false alarms under these cost assumptions.

Drift: When we trained a Random Forest on the first half of the data (by time) and tested it on the second half, PR-AUC fell from 0.795 (random split) to 0.754 (chronological split) - about a 5% relative decline. This confirms that fraud patterns shift even within a two-day window, and any production deployment of this model should include a **scheduled retraining pipeline** rather than treating the model as "finished" after one training run.

Explainability: SHAP analysis on the XGBoost model showed 'V14', 'V4', 'V11', 'V12', and 'V1' as the five most influential features in fraud predictions - these are the signals a fraud analyst would want to monitor most closely for future shifts.

Final Recommendation: We recommend deploying the **XGBoost** model, balanced using **SMOTETomek**, scored using the cost-sensitive threshold of **0.12** rather than the default 0.5, and monitored on a regular schedule for drift with periodic retraining as new confirmed fraud/genuine labels become available.